FACULTY OF COMPUTING
UTM Johor Bahru

SECP3213: BUSINESS INTELLIGENCE

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SECTION 1

PROJECT PHASE 4:

Final Report

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1 PROJECT TITLE

Analyzing the Impact of Logistics Performance on Customer Satisfaction and Revenue Trends in Brazilian E-Commerce

2 PROJECT BACKGROUND

The e-commerce industry requires a comprehensive understanding of the entire transaction lifecycle, encompassing customer actions, order processing, and logistics. Operating within the Brazilian market, platforms like Olist must manage complex interactions involving diverse customer locations, specific product catalogs, dynamic shipping rates, and exact geolocation coordinates for buyers and sellers (Olist et al., 2021).

Operational efficiency is heavily reliant on delivery performance, yet it can be difficult to measure how logistical bottlenecks such as prolonged shipping durations or late deliveries directly influence negative customer reviews and overall satisfaction. Additionally, businesses face challenges in accurately tracking seasonal purchasing behaviors and identifying which specific geographical regions are driving sales volumes versus those experiencing delivery delays.

By implementing a two-tier Business Intelligence architecture using Alteryx Designer for ETL processes (Alteryx, 2026) and Tableau or Power BI for analytics (Microsoft, 2025), raw fragmented data can be transformed into a cohesive analytical model. Developing interactive dashboards will empower users to explore sales trends dynamically, evaluate the operational impact on customer satisfaction, and make data-driven decisions to optimize regional logistics.

3 PROJECT OBJECTIVES

This project pursues five principal objectives. First, it investigates and visualizes the correlation between logistics performance, particularly prolonged shipping durations, and customer satisfaction as reflected in review scores. Second, it analyses seasonal purchasing behaviours and tracks monthly and quarterly revenue trends and spikes across the platform. Third, it performs geographic segmentation to identify which Brazilian states generate the highest sales volume and compares them against states experiencing the longest delivery delays. Fourth, it identifies the top-performing product categories and evaluates the specific sellers driving the highest sales volume across the platform. Fifth, it designs and executes an automated Extract, Transform, Load (ETL) workflow using Alteryx Designer to clean, standardize, and integrate five fragmented e-commerce datasets into a single cohesive data model.

4 DATASET DESCRIPTION

This project utilises five interconnected datasets sourced from the Brazilian E-Commerce Public Dataset by Olist, available on Kaggle (Olist et al., 2021). Together, these datasets provide a holistic representation of e-commerce operations in Brazil, encompassing customer behaviour, order processing, and logistics.

4.1 Customers Dataset

The Customers Dataset identifies unique customers and their physical locations across Brazil. It serves as the foundation for tracking repeat purchase behaviour and conducting regional demographic analysis to deepen understanding of the platform's customer base. The dataset is structured in CSV format and is sourced from the Brazilian E-Commerce Public Dataset by Olist, publicly available on Kaggle. Table 1 presents the key variables contained within this dataset along with their respective descriptions.

Table 1: Key Variables of the Customers Dataset

Key Variables	Description
customer_id	Unique identifier that links to orders dataset.
customer_unique_id	Unique identifier for a customer across multiple orders
customer_zip_code_prefix	First five digits of the customer's zip code.
customer_city	Name of the city where the customer is located.
customer_state	State abbreviation such as “SP” or “RJ”.

4.2 Orders Dataset

As the core dataset, it records the complete lifecycle and status of every transaction made on the platform. It shows critical timestamps for purchase, approval, and delivery, which are used to measure operational efficiency. This data is the main source for analysing delivery performance and order fulfilment trends. The dataset is structured in CSV format and is sourced from the Brazilian E-Commerce Public Dataset by Olist, publicly available on Kaggle. Table 2 presents the key variables contained within this dataset along with their respective descriptions.

Table 2: Key Variables of the Orders Dataset

Key Variables	**Description**
order_id	Unique identifier for each order.
customer_id	Link to the customer dataset.
order_status	Status for each order.
order_purchase_timestamp	Date and time when the purchase was made.
order_delivered_customer_date	Actual delivery date to the customer.
order_estimated_delivery_date	Estimated delivery date promised at the time of purchase.

4.3 Orders Items Dataset

Orders Items dataset shows a full view of every product sold, including its price and associated freight costs. It allows for detailed analysis and identifies which sellers and products drive the most volume. Besides that, it also acts as the bridge between order transactions and specific product catalogue details. The dataset is structured in CSV format and is sourced from the Brazilian E-Commerce Public Dataset by Olist, publicly available on Kaggle. Table 3 presents the key variables contained within this dataset along with their respective descriptions.

Table 3: Key Variables of the Orders Items Dataset

Key Variables	**Description**
order_id	Link to the orders dataset.
product_id	Unique identifier for the product.
seller_id	Unique identifier for the seller.
price	Item price for each unit.
freight_value	Shipping cost for the item.
shipping_limit_date	Seller's deadline for handling the order.

4.4 Order Reviews Dataset

The Order Reviews dataset captures customer feedback and satisfaction scores submitted following the completion of each transaction. It contains both numerical ratings and free-text comments, enabling service quality analysis across a range of dimensions. By linking review data to order records, it becomes possible to examine whether logistical issues such as delayed deliveries have a measurable effect on customer satisfaction. The dataset is structured in CSV format and is sourced from the Brazilian E-Commerce Public Dataset by Olist, publicly available on Kaggle. Table 4 presents the key variables contained within this dataset along with their respective descriptions.

Table 4: Key Variables of the Orders Reviews Dataset

Key Variables	**Description**
review_id	Unique identifier for the review.
order_id	Link to the orders dataset.
review_score	Rating ranging from 1 to 5.
review_comment_message	Text content of the customer's review. It is an unstructured component within a structured file

review_creation_date	Date the survey was sent.

4.5 Geolocation Dataset

The Geolocation dataset gives exact latitude and longitude data for Brazilian zip codes that are utilized by buyers and sellers. It is useful in determining shipping distances, as well as developing geospatial maps including sales heat maps. The dataset is structured in CSV format and is sourced from the Brazilian E-Commerce Public Dataset by Olist, publicly available on Kaggle. Table 5 presents the key variables contained within this dataset along with their respective descriptions.

Table 5: Key Variables of the Geolocation Dataset

Key Variables	Description
geolocation_zip_code_prefix	Link to customer/seller zip codes.
geolocation_lat	Latitude coordinates.
geolocation_lng	Longitude coordinates.
geolocation_city	Name of the city.
geolocation_state	State abbreviation.

5 PROPOSED BUSINESS INTELLIGENCE APPROACH

This chapter outlines the proposed Business Intelligence (BI) solution designed to address the analytical objectives established in the preceding sections. It details the tools selected for data processing and visualisation, the structured workflow adopted for data preparation, and the analytical and visualisation strategies employed to derive meaningful insights from the dataset. Collectively, these components form an end-to-end BI pipeline, ranging from raw data ingestion through to interactive dashboard delivery, intended to support data-driven decision-making across key business dimensions including sales performance, logistics efficiency, and customer satisfaction.

5.1 Tool Utilization Strategy

The proposed Business Intelligence solution uses a two-tier architecture, as summarised in Table 6, integrating Alteryx Designer as the backend data preparation engine with a front-end visualization platform, either Tableau or Microsoft Power BI, to deliver a cohesive and interactive analytical environment.

Alteryx Designer will serve as the core data engine that is responsible for all Extract, Transform, and Load (ETL) processes. It will be used to ingest the raw CSV files specifically the orders, items, reviews, and customer datasets. We will use Alteryx Designer to standardize the formats and blend all of it into a single cohesive relational dataset. For the last step, the necessary aggregations are performed to ready the data for visual analysis.

Once the data is thoroughly cleaned and prepped, it will be imported into a visualization tool, such as Tableau or Microsoft Power BI. Within this front-end platform, we will build interactive dashboards featuring dynamic slicers such as date ranges and geographic regions that empower stakeholders to intuitively explore sales trends as well as pinpoint logistics bottlenecks and analyse customer satisfaction metrics.

Table 6: Summary of BI Tools and Their Respective Purposes

Tools Used	**Purpose**	**Description**
Alteryx Designer	To create data workflow and ETL (Extract, Transform, Load)	Act as the data engine. It will ingest the raw CSV files and standardize the formats which will be combined into a single cohesive dataset. Aggregations can be performed upon the dataset for visual analysis.
Tableau / Power BI	To create dashboard with interactions	The cleaned dataset from Alteryx Designer will be imported to be used for creation of interactive dashboards with slicers to allow

		users to explore sales trends and customer satisfaction.

5.2 Data Workflow

To ensure accurate and meaningful analysis, all datasets will undergo a rigorous structured preparation process within Alteryx Designer, executed across three distinct phases as outlined in Table 7.

Table 7: Data Workflow Phases, Tools, and Methodology

Phase	Tools	Methodology
Data Integration	Join Tool	• Merge the Orders, Items, and Reviews datasets using the common order_id key. • Join the Customers dataset using customer_id to link geographic data to each transaction.
Data Cleaning	Data Cleansing Filter Tool DateTime	• Filter out incomplete records and remove duplicate rows. • Convert string-based timestamps into actual date formats for accurate time-series analysis.
Data Transformation	Formula Tool Summarize Tool	• Calculate Shipping Duration • Group data by customer state and product category to calculate total revenue, average shipping times, and satisfaction scores.

5.3 Analysis and Visualisation Strategy

The project will focus on identifying correlations between logistics performance and customer satisfaction, as well as tracking revenue trends across different regions. To thoroughly understand the business problem, our team will conduct three primary forms of investigation. First, we will perform a trend analysis to track seasonal purchasing behaviours and identify any monthly revenue spikes over the dataset's timeframe.

Next, we will execute geographic segmentation to determine which Brazilian states drive the highest sales volumes and which specific regions struggle with the longest delivery delays. Finally, we will run a correlation analysis to explicitly investigate the relationship between prolonged shipping durations and negative customer review scores. Appropriate Visual Forms effectively communicate these insights to stakeholders. We will leverage five specific visual formats in our final dashboard. We will use a line chart to clearly visualize revenue trends and order volumes over monthly and quarterly timeframes.

A geospatial map will be implemented to display the concentration of customers and total sales distributions across various Brazilian states. To illustrate our correlation findings, a scatter plot will map Shipping Duration on the x-axis against Customer Review Score on the y-axis. We will also rely on a bar chart to compare the top 10 best-selling product categories or rank the states experiencing the highest average delivery times.

Finally, we will feature KPI text cards to instantly highlight high-level metrics briefly, specifically focusing on Total Revenue, Average Delivery Time and Average Satisfaction Score.

6 WORKFLOW OVERVIEW & TOOLS USED

As illustrated in Figure 1, the Business Intelligence (BI) architecture was developed using Alteryx Designer to construct an end-to-end analytical workflow. Various Alteryx tools were employed across different stages of the data processing pipeline. For data ingestion and export, the Data Input and Data Output tools were used to facilitate seamless data access and storage. During the data preparation phase, tools such as Filter, Data Cleansing, Unique, Select, and Sort were applied to remove inconsistencies, handle missing values, eliminate duplicate records, and organize the dataset for analysis. Data transformation and integration were performed using the Join, Formula, and Summarize tools, enabling the combination of multiple datasets, creation of derived attributes, and aggregation of key metrics. Finally, for data investigation and analytical purposes, the Pearson Correlation tool was utilized to examine the strength and direction of relationships between selected variables, providing insights to support data-driven decision-making within the BI framework.

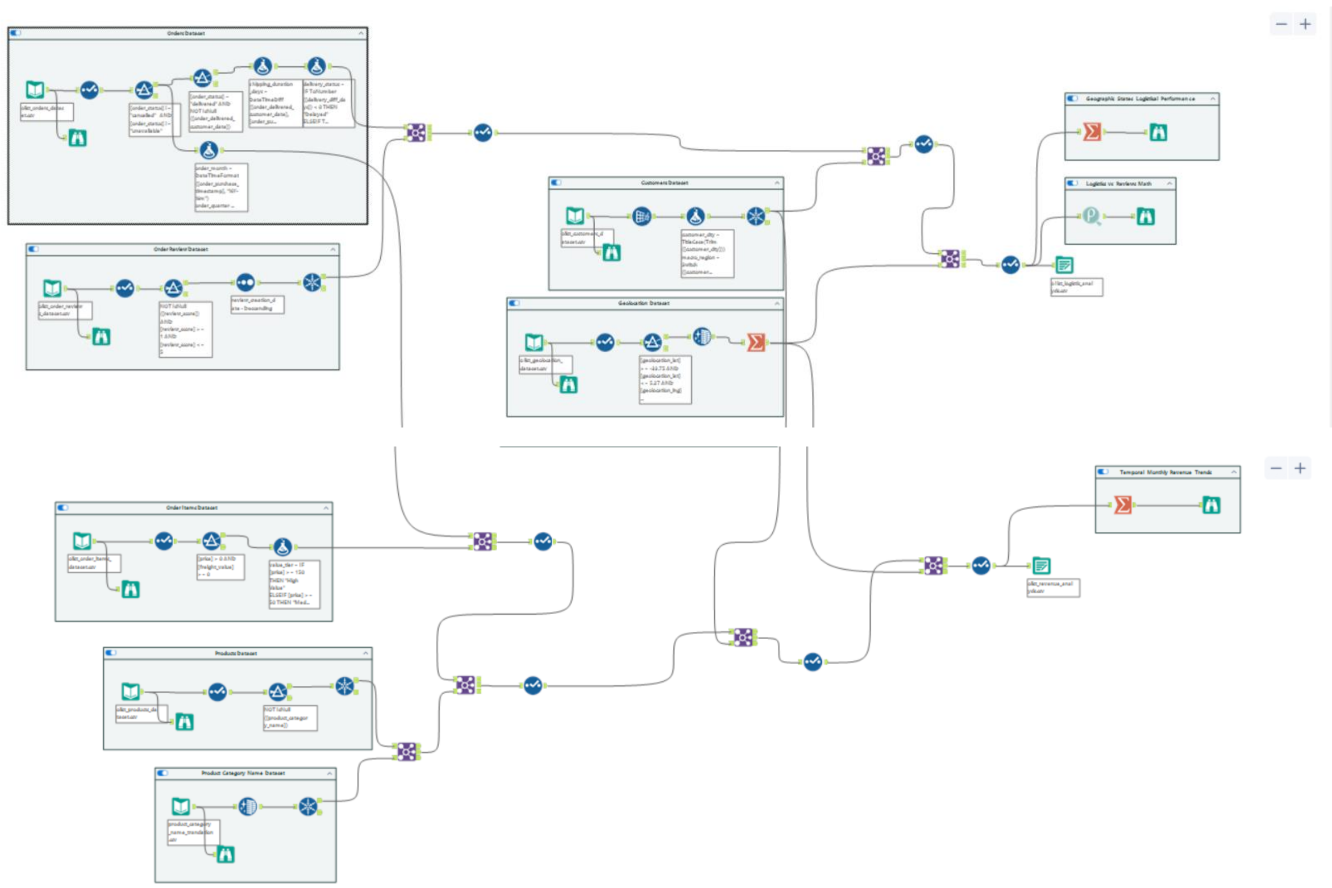
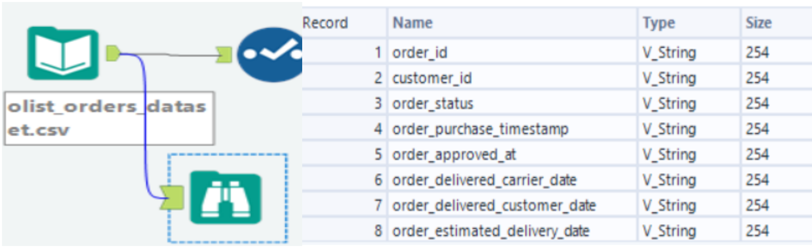


Figure 1: Overview of the Alteryx Workflow for the Business Intelligence Architecture.

7 DATA INPUT FROM MULTIPLE DATASETS

Our workflow ingested seven distinct CSV datasets from the Olist Brazilian E-Commerce database (Olist et al., 2021). This was executed using multiple Input Data tools configured to read comma-delimited files.

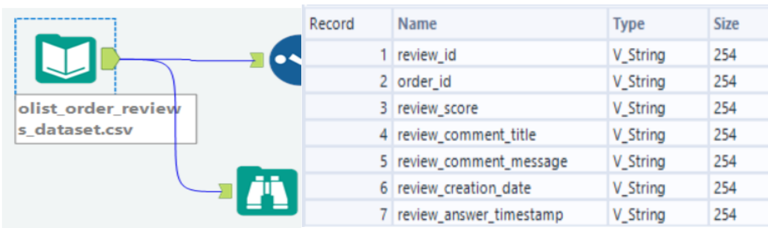
1. **Orders Dataset (olist_orders_dataset.csv):** Contains core transaction statuses and purchase timestamps, as illustrated in Figure 2



Record	Name	Type	Size
1	order_id	V_String	254
2	customer_id	V_String	254
3	order_status	V_String	254
4	order_purchase_timestamp	V_String	254
5	order_approved_at	V_String	254
6	order_delivered_carrier_date	V_String	254
7	order_delivered_customer_date	V_String	254
8	order_estimated_delivery_date	V_String	254

Figure 2 : Orders Dataset – Schema and Field Types

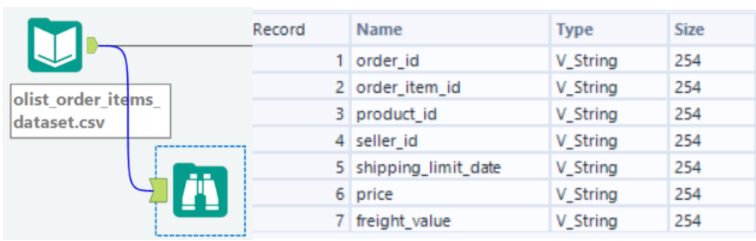
2. **Order Reviews Dataset (olist_order_reviews_dataset.csv):** Captures post-purchase customer feedback, specifically the 1-to-5 satisfaction scores and timestamped review dates, as illustrated in Figure 3.



Record	Name	Type	Size
1	review_id	V_String	254
2	order_id	V_String	254
3	review_score	V_String	254
4	review_comment_title	V_String	254
5	review_comment_message	V_String	254
6	review_creation_date	V_String	254
7	review_answer_timestamp	V_String	254

Figure 3: Order Reviews Dataset – Schema and Field Types

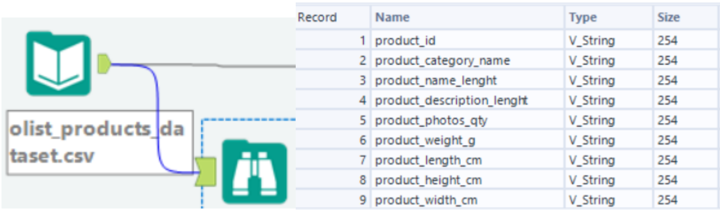
3. **Order Items Dataset (olist_order_items_dataset.csv):** Contains the line-item financial details for each transaction, including product IDs, item prices, and freight values, as illustrated in Figure 4



Record	Name	Type	Size
1	order_id	V_String	254
2	order_item_id	V_String	254
3	product_id	V_String	254
4	seller_id	V_String	254
5	shipping_limit_date	V_String	254
6	price	V_String	254
7	freight_value	V_String	254

Figure 4: Order Items Dataset – Schema and Field Types

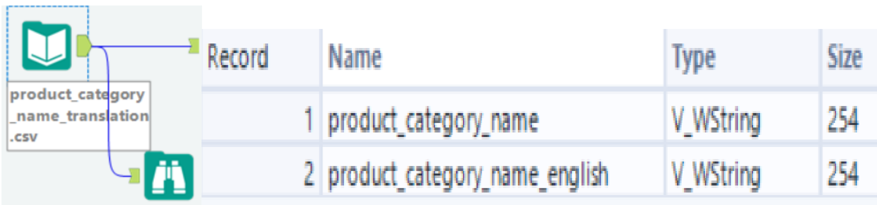
4. **Products Dataset (olist_products_dataset.csv):** Contains physical product attributes and assigns a Portuguese product category name to each item ID.



Record	Name	Type	Size
1	product_id	V_String	254
2	product_category_name	V_String	254
3	product_name_lenght	V_String	254
4	product_description_lenght	V_String	254
5	product_photos_qty	V_String	254
6	product_weight_g	V_String	254
7	product_length_cm	V_String	254
8	product_height_cm	V_String	254
9	product_width_cm	V_String	254

Figure 5: Products Dataset – Schema and Field Types

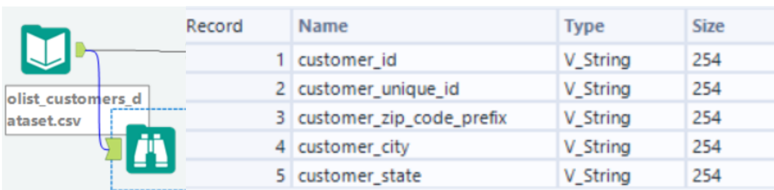
5. **Product Category Name Translation (product_category_name_translation.csv):** A mapping table used to seamlessly translate the Portuguese product category names into English as illustrated in Figure 6, ensuring the final dashboards are readable for English-speaking stakeholders



Record	Name	Type	Size
1	product_category_name	V_WString	254
2	product_category_name_english	V_WString	254

Figure 6: Product Category Name Translation Dataset – Schema and Field Types

6. **Customers Dataset (olist_customers_dataset.csv):** Provides demographic mapping, linking unique customer IDs to their geographic locations which includes city, state, and zip code prefix as illustrated in Figure 7.



Record	Name	Type	Size
1	customer_id	V_String	254
2	customer_unique_id	V_String	254
3	customer_zip_code_prefix	V_String	254
4	customer_city	V_String	254
5	customer_state	V_String	254

Figure 7: Customers Dataset – Schema and Field Types

7. **Geolocation Dataset (olist_geolocation_dataset.csv):** A highly granular spatial lookup table mapping Brazilian zip codes to specific latitude and longitude coordinates as illustrated in Figure 8.

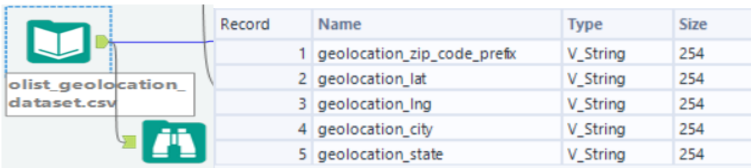


Figure 8: Geolocation Dataset – Schema and Field Types

7.1 Data Cleaning

To ensure the integrity and analytical readiness of the ingested datasets, a series of data cleaning operations were applied across multiple tools within the Alteryx Designer workflow.

7.1.1 Select Tool

A Select Tool was applied immediately following the Input Data tool for the Orders Dataset to correct the automatic data type detection. All five timestamp columns, which had been read as V_String by default, were manually reconfigured to the DateTime data type to enable date-based calculations in downstream processing, as shown in Figure 9.

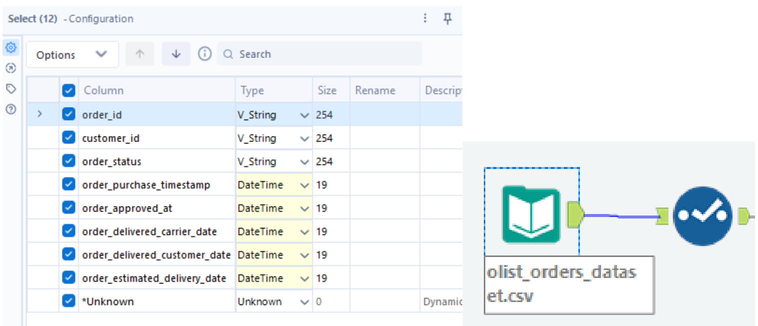


Figure 9: Select Tool Configuration – Timestamp Data Type Correction

7.1.2 Filter Tool

A second Filter Tool was applied to split the cleaned orders dataset into two analysis streams, as shown in Figure 10. The True output produces Stream A (logistic analysis) containing only fully delivered orders with confirmed customer delivery dates, which is used for logistics performance and customer satisfaction analysis. The False output produces Stream B (revenue analysis) containing all remaining non-cancelled orders regardless of delivery status, which is used for revenue trend and product category analysis.

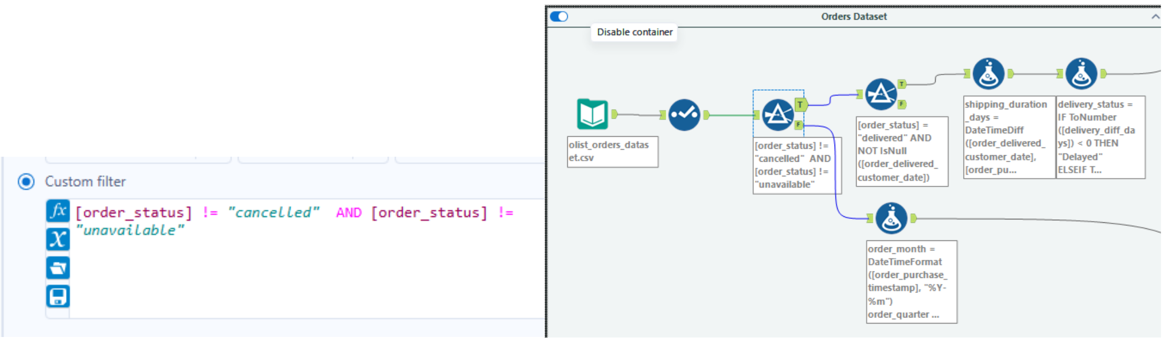


Figure 10: Filter Tool Configuration – Stream A and Stream B Partitioning

7.1.3 Data Cleanse Tool

A Data Cleanse Tool was applied to the geolocation dataset to remove leading and trailing whitespace from the geolocation_city and geolocation_state columns, as illustrated in Figure 11.

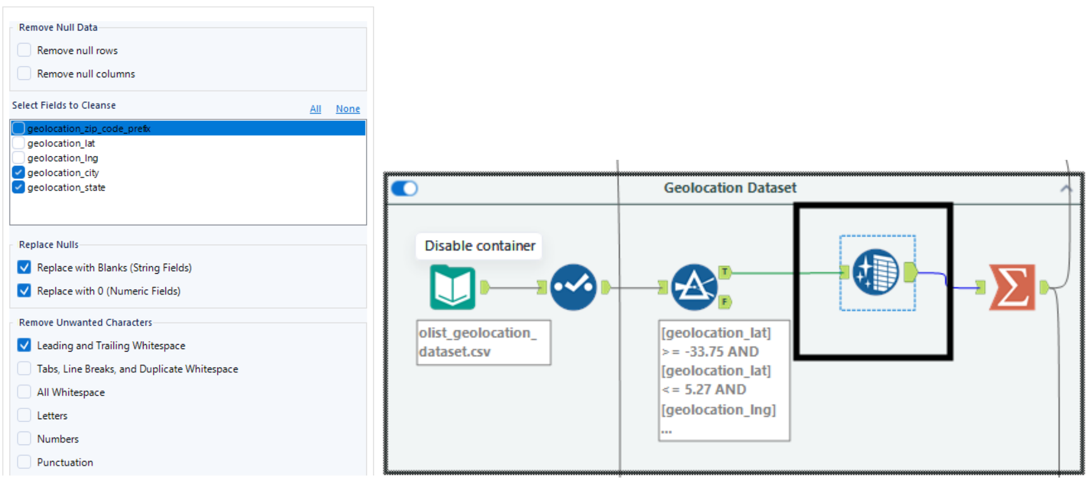


Figure 11: Data Cleanse Tool Configuration – Geolocation Dataset

7.1.4 Unique Tool

A Unique Tool was applied to the customers dataset to remove duplicate customer records based on the customer_unique_id field as illustrated in Figure 12.

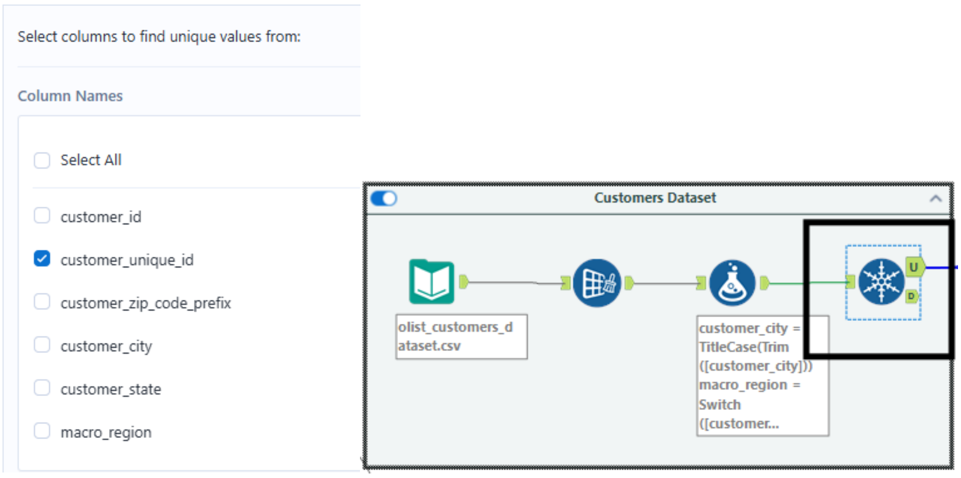


Figure 12: [Figure 6: Unique Tool Configuration – Customers Dataset]

7.1.5 Filter Tool

A Filter Tool was applied to the geolocation dataset to remove coordinate entries that fall outside Brazil's valid geographic boundaries as illustrated in Figure 13.

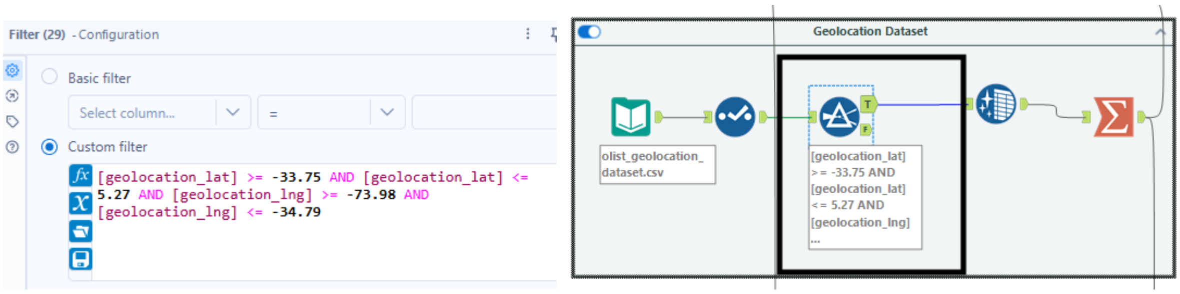


Figure 13: Filter Tool Configuration – Geolocation Boundary Filtering

7.2 Data Transformation

Following the data cleaning phase, a series of transformation operations were performed to derive new analytical fields and aggregate the cleaned datasets into summary outputs suitable for dashboard consumption.

7.2.1 Formula Tool – Order Month and Year Derivation

The order_month field is derived using the DateTimeFormat function with the format pattern '%Y-%m', producing a year-month value such as '2018-03' that enables monthly revenue aggregation as illustrated in Figure 14.

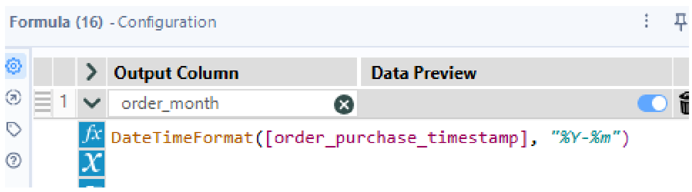


Figure 14: Formula Tool Configuration – Order Month Derivation

7.2.2 Summarise Tool – State-Level Logistics Aggregation

A summarize tool is configured to aggregate logistics performance metrics at the state level for geographic analysis in Objective 3. The tool groups all records by customer_state and calculates the average shipping_duration_days, average delivery_diff_days, and total order count per state. Average latitude and longitude values are also retained to support geospatial plotting in the dashboard, as shown in Figure 15.

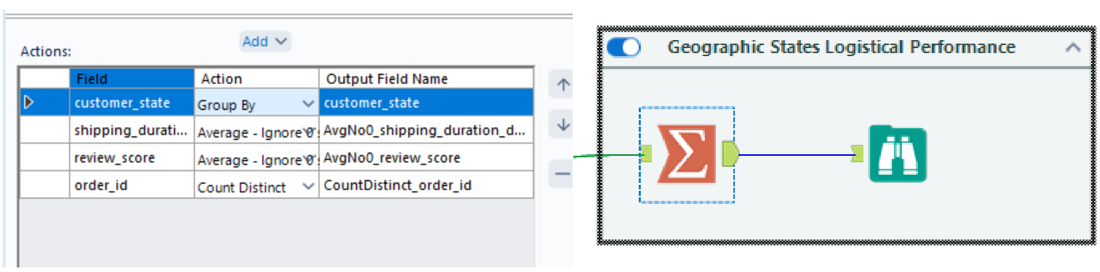


Figure 15: Summarise Tool Configuration – Geographic Logistics Performance Aggregation

7.2.3 Summarise Tool – Geolocation Aggregation

The geolocation dataset contains many duplicate entries for the same zip code, each with slightly different coordinate values. A Summarize Tool was applied to collapse all these duplicate zip code entries into one single representative row per zip code as shown in Figure 16.

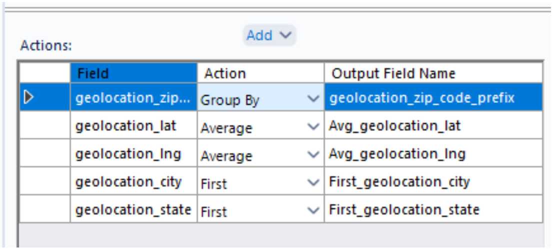


Figure 16: Summarise Tool Configuration – Geolocation Deduplication

7.3 Data Integration

Data integration was executed primarily through the Alteryx Join Tool across both analysis streams. Following each Join operation, a Select Tool was immediately applied to deselect duplicate key columns generated automatically by the Join Tool, such as Right_order_id and Right_customer_id, to optimise processing efficiency and maintain a clean dataset structure.

7.3.1 The Transactional Dimension – Orders and Items

The core of the workflow was established by joining the *Orders* dataset which contains delivery timelines, with the *Order Items* dataset which contains financial pricing, using the order_id primary key. This served as the core transactional workflow, as illustrated in Figure 17.

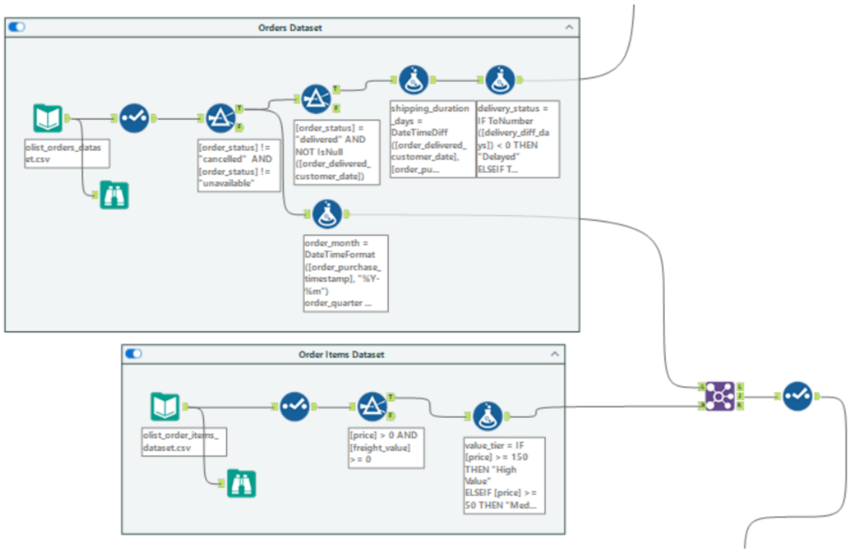


Figure 17: Join Tool Configuration – Orders and Order Items Datasets

7.3.2 The Customer and Spatial Dimension

To physically map where logistical bottlenecks occur, the spine was joined with the *Customers* dataset via customer_id. This enriched stream was then joined to the aggregated *Geolocation* dataset via zip_code_prefix to attach accurate latitude and longitude coordinates to every individual order, as shown in Figure 18.

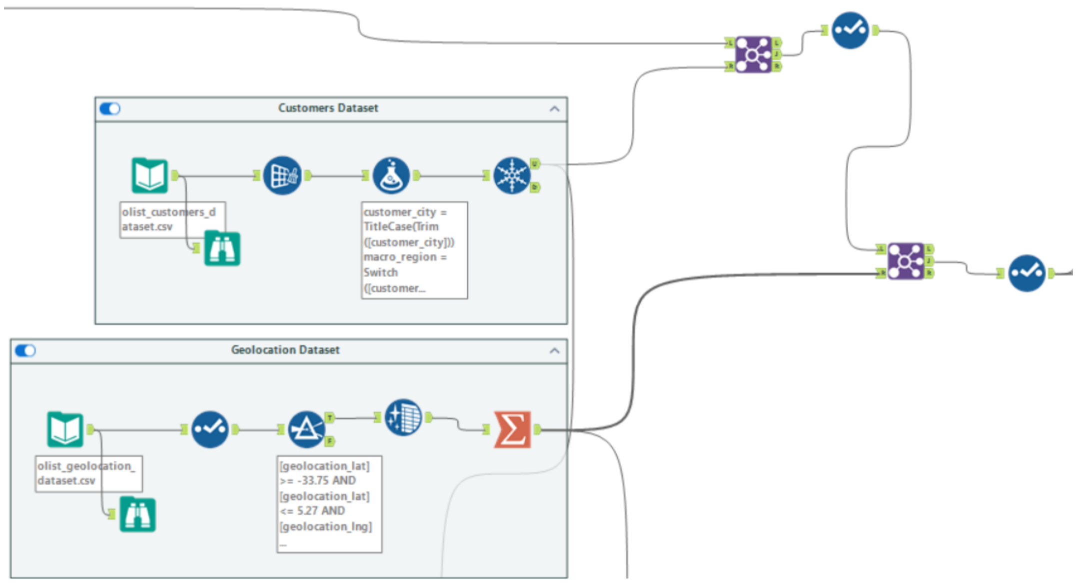


Figure 18: Join Tool Configuration – Customer and Geolocation Enrichment

7.3.3 The Product Dimensions

The *Products* dataset was integrated using product_id to attach physical item attributes. A secondary join was then executed with the *Product Category Translation* table on product_category_name to systematically convert the native Portuguese data into English to ensure dashboard readability for stakeholders, as shown in Figure 19.

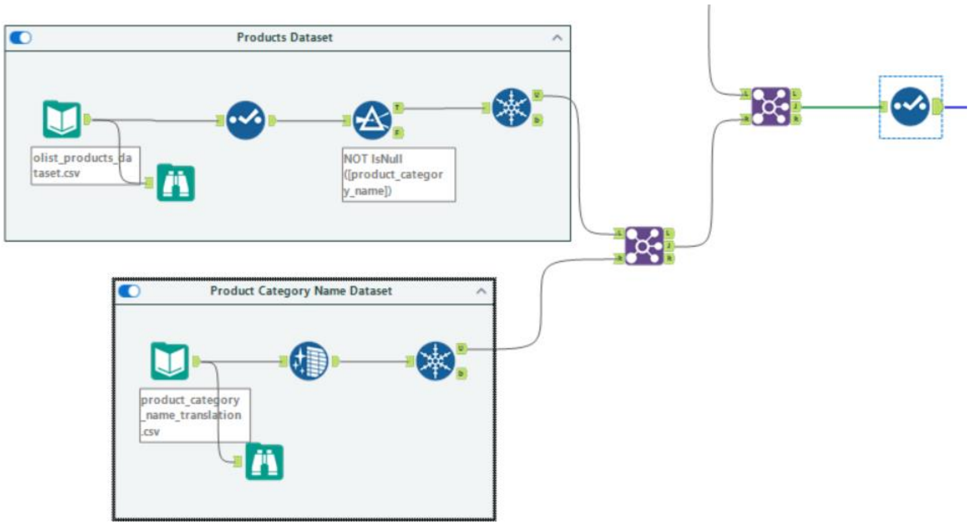


Figure 19: Join Tool Configuration – Product and Category Name Translation

7.3.4 The Satisfaction Dimension

The *Order Reviews* dataset was joined via order_id. This was the critical final link, attaching 1-to-5 star customer satisfaction scores directly to the logistic delivery variables, setting up the foundation for our predictive analytics, as shown in Figure 20.

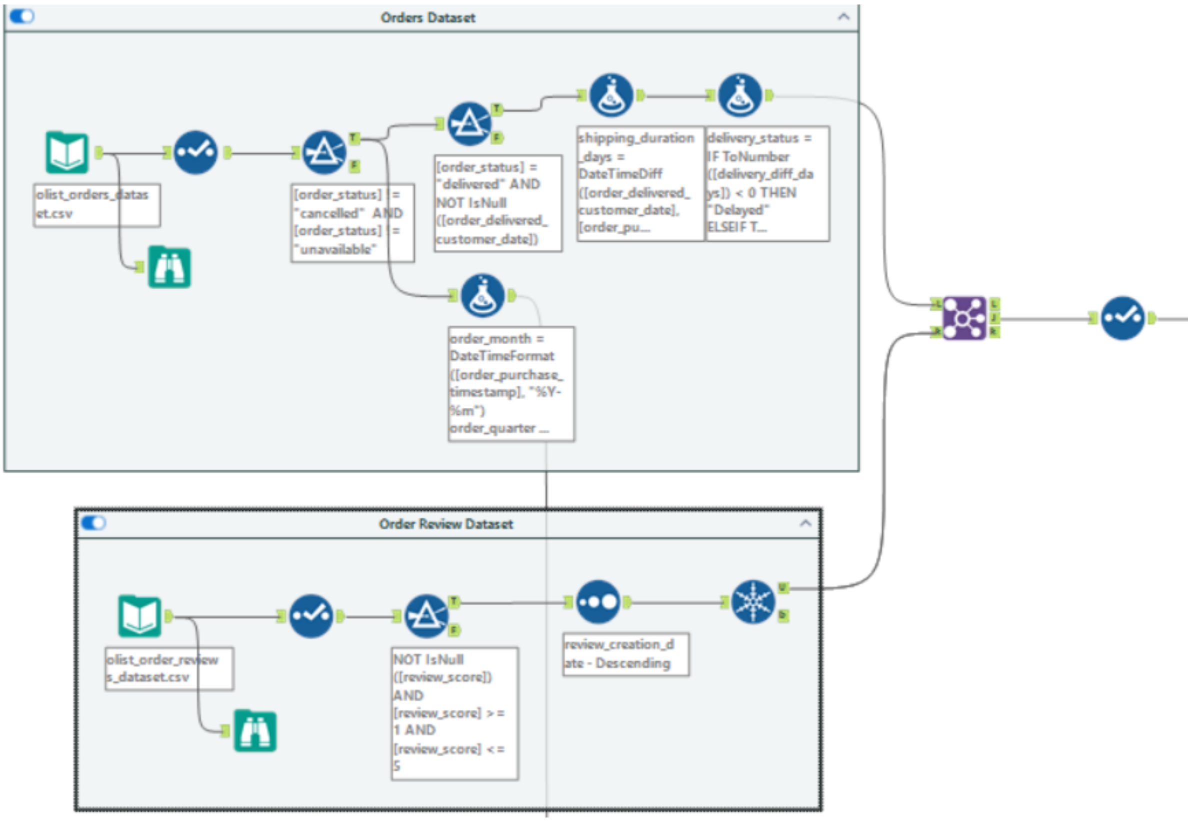


Figure 20: Join Tool Configuration – Order Reviews Integration

7.4 Feature Creation

Beyond the core transformation steps, three derived features were engineered to enrich the dataset with business-relevant categorical variables.

7.4.1 Delivery Status Classification

A Formula Tool was applied to convert raw shipping day calculations into a categorical delivery status field using conditional IF/ELSE logic. Orders where delivery_diff_days is less than zero are classified as "Early," orders where the value equals zero are classified as "On Time," and all remaining orders are classified as "Delayed." This segmentation directly enables the correlation of late deliveries with negative customer review scores, as shown in Figure 21.

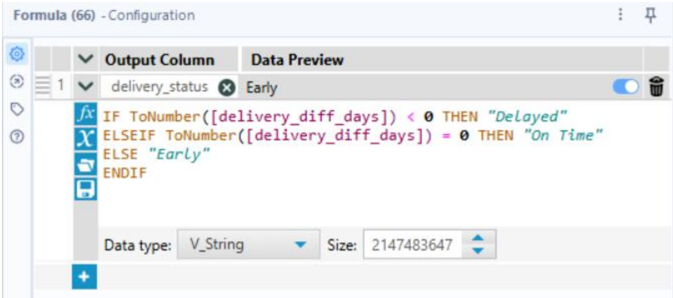


Figure 21: Formula Tool Configuration – Delivery Status Classification

7.4.2 Customer Value Tier Segmentation

Individual transaction prices within the Order Items Dataset were segmented into three customer value tiers using a Formula Tool. Transactions with a price of $150 or above are classified as "High Value," those at $50 or above are classified as "Medium Value," and all remaining transactions are classified as "Low Value," as depicted in Figure 22.

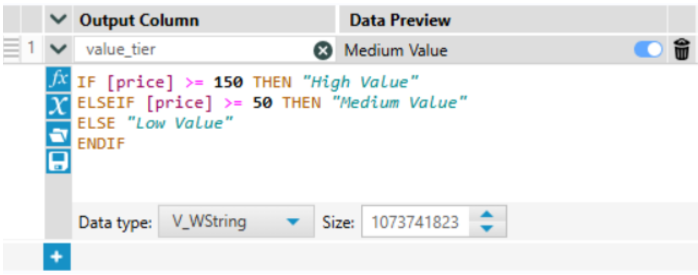


Figure 22: Formula Tool Configuration – Customer Value Tier Segmentation

7.4.3 Geospatial Macro-Region Classification

To solve visual clutter problem from the mapping of all 27 distinct Brazilian states individually, a Switch function was applied within a Formula Tool to consolidate the states into five higher-level macro-regions, namely Southeast, South, North, Northeast, and Center-West. This grouping produces a cleaner and more interpretable geographic segmentation for dashboard visualisation, as illustrated in Figure 23.

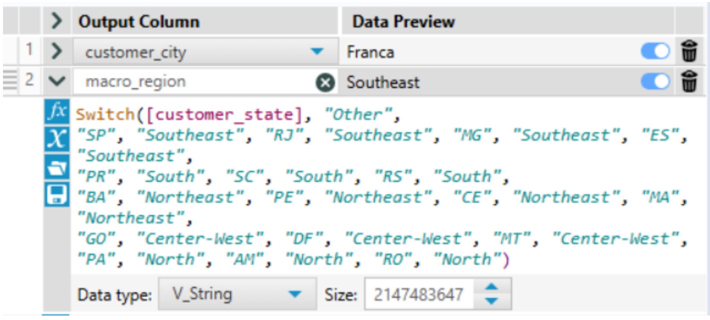


Figure 23: Formula Tool Configuration – Macro-Region Classification

8 ANALYTICAL TECHNIQUE AND RESULTS

Three core analytical methods were integrated within the Alteryx workflow to extract measurable business intelligence value prior to exporting the processed datasets to their destination output files.

8.1 Pearson Correlation Analysis

A Pearson Correlation Tool was applied to the completed Logistics Analysis stream to quantify the relationship between shipping intervals and customer satisfaction scores. The tool was configured with shipping_duration_days and review_score as the two input variables. The analysis yielded a correlation coefficient (r) of -0.3357, as shown in Figure 24.

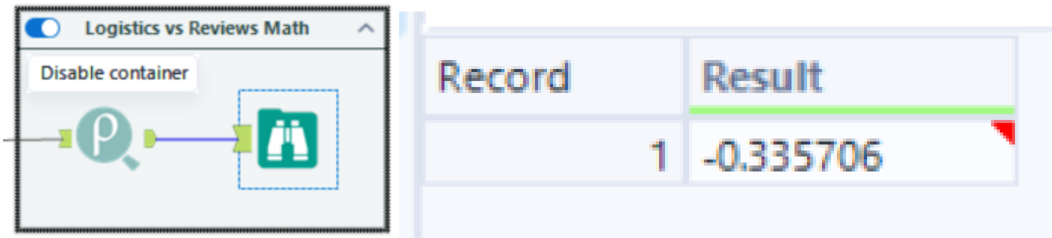
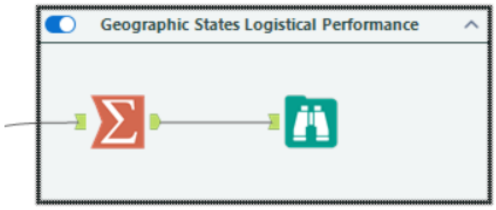


Figure 24: Pearson Correlation Tool Output – Shipping Duration vs. Review Score

This result confirms a moderate negative linear correlation between the two variables, providing mathematical evidence that as shipping durations increase, customer review scores decline in a statistically predictable manner. This finding establishes an empirical baseline to support logistics optimisation recommendations.

8.2 Geographic Segmentation Matrix

A Summarise Tool was attached to the Logistics Analysis stream to aggregate operational performance metrics across regional segments. The tool was configured to group all records by customer_state and to compute the average shipping_duration_days and average review_score per state. The resulting output is a structured matrix evaluating logistics bottlenecks and satisfaction performance grades across all 27 Brazilian regions, as depicted in Figure 25.



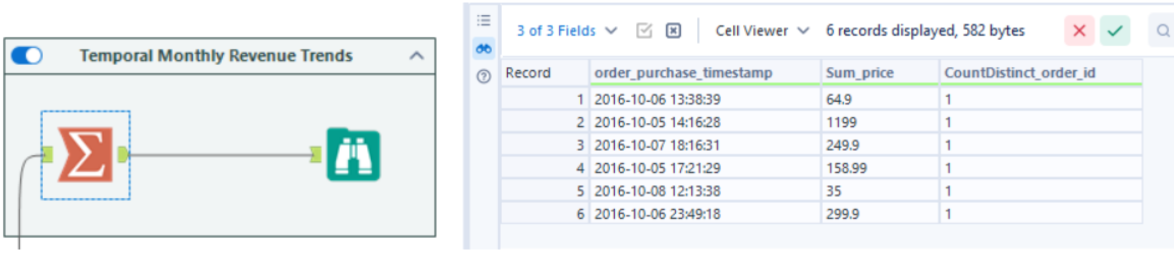
4 of 4 Fields | 27 records displayed, 1,282 bytes

Record	customer_state	AvgNo0_shipping_duration_days	AvgNo0_review_score	CountDistinct_order_id
1	MS	15.200886	4.162482	677
2	MT	17.463443	4.147406	848
3	TO	17.203774	4.169811	265
4	ES	15.179663	4.065859	1898
5	PA	23.203969	3.9129	907
6	PR	11.503593	4.238377	4732
7	AP	26.815385	4.246154	65
8	RR	29.475	3.875	40
9	SC	14.45183	4.133821	3415
10	SP	8.285119	4.245509	38858
11	PB	19.843058	4.064386	497
12	SE	20.96	3.895385	325
13	PI	19.096491	3.986842	456
14	RO	18.69469	4.199115	226
15	AC	19.776316	4.197368	76
16	RS	14.779961	4.185798	5140
17	DF	12.412338	4.133658	1848
18	AM	26.172662	4.230216	139
19	PE	17.92212	4.073953	1528
20	RN	18.825328	4.137555	458
21	AL	24.013123	3.860892	381
22	RJ	14.776095	3.962998	11756
23	GO	15.105882	4.100535	1870
24	MG	11.511911	4.190398	10914
25	MA	20.947826	3.826087	690
26	BA	18.79686	3.922486	3122
27	CE	20.862871	3.943865	1247

Figure 25: Geographic States Logistical Performance Output Matrix

8.3 Temporal Trend Analysis Matrix

A secondary Summarise Tool was connected to the Revenue Analysis stream to evaluate historical business growth trends over time. The tool was configured to group records by the derived order_month field and to calculate the sum of payment_value alongside a distinct count of order_id. The resulting time-series matrix measures transactional demand and commercial revenue velocity across the full duration of Olist's operating history, as illustrated in Figure 26.



Record	order_purchase_timestamp	Sum_price	CountDistinct_order_id
1	2016-10-06 13:38:39	64.9	1
2	2016-10-05 14:16:28	1199	1
3	2016-10-07 18:16:31	249.9	1
4	2016-10-05 17:21:29	158.99	1
5	2016-10-08 12:13:38	35	1
6	2016-10-06 23:49:18	299.9	1

Figure 26: Temporal Monthly Revenue Trends Output Matrix

9 CHALLENGES FACED

The development of the Alteryx workflow encountered several technical and structural challenges that required deliberate resolution strategies. The first challenge involved incorrect data type detection for timestamp columns. All five timestamp fields within the Orders Dataset were automatically interpreted as V_String by Alteryx's default input configuration, which prevented any date-based arithmetic from being performed. This was resolved by manually reconfiguring the affected columns to the DateTime data type using the Select Tool immediately following data ingestion.

The second challenge concerned duplicate zip code entries within the Geolocation Dataset. The dataset contained numerous duplicate records sharing the same zip code but carrying marginally differing coordinate values. This was addressed by applying a Summarise Tool to collapse all duplicate entries into a single representative row per zip code using averaged coordinate values.

The third challenge involved invalid coordinate entries within the Geolocation Dataset. Certain records contained latitude and longitude values that fell outside Brazil's valid geographic boundaries, which would have resulted in incorrectly plotted data points on geospatial visualisations. A Filter Tool was applied to retain only coordinate records within Brazil's true geographic range, removing all erroneous entries from the dataset.

The fourth challenge arose from the storage of all product category names in Portuguese. As these labels would have rendered unreadable on stakeholder-facing dashboards, an additional

Product Category Name Translation dataset was incorporated and joined to the Products Dataset. This introduced an extra integration step and necessitated careful post-join column management to prevent duplication.

The fifth challenge was the automatic generation of duplicate key columns following every Join operation. The Alteryx Join Tool consistently produced columns prefixed with "Right_" for each shared key field, which accumulated across the sequential join chain and inflated the dataset unnecessarily. A Select Tool was placed immediately after every Join Tool throughout both streams to drop these redundant columns and preserve a clean, minimal dataset structure.

The sixth and final challenge involved managing two parallel analysis streams concurrently within a single workflow. The True and False output anchors of the Filter Tool were used to route data into Stream A and Stream B respectively. Ensuring that both streams correctly integrated with shared reference datasets, namely the Customers and Geolocation datasets, without introducing conflicts required careful workflow planning and deliberate canvas organisation throughout the design process.

10 SYSTEM ARCHITECTURE & DESIGN FRAMEWORK

To bridge backend data preparation and descriptive business analytics, the processed datasets from Alteryx (olist_logistic_analysis.csv and olist_revenue_analysis.csv) were connected directly to Microsoft Power BI Desktop. The visualization layer was architected into a cohesive system across three distinct, user-filtered dashboard viewpoints designed to translate data distributions into commercial action items, adhering to dashboard interface frameworks established by Few (2013) and Microsoft (2025).

11 DASHBOARD PAGE 1: BUSINESS & PRODUCT ANALYSIS OVERVIEW

This dashboard serves as the central hub for macro-level corporate tracking, monitoring high-level KPIs, customer spending brackets, and product category financial performance.

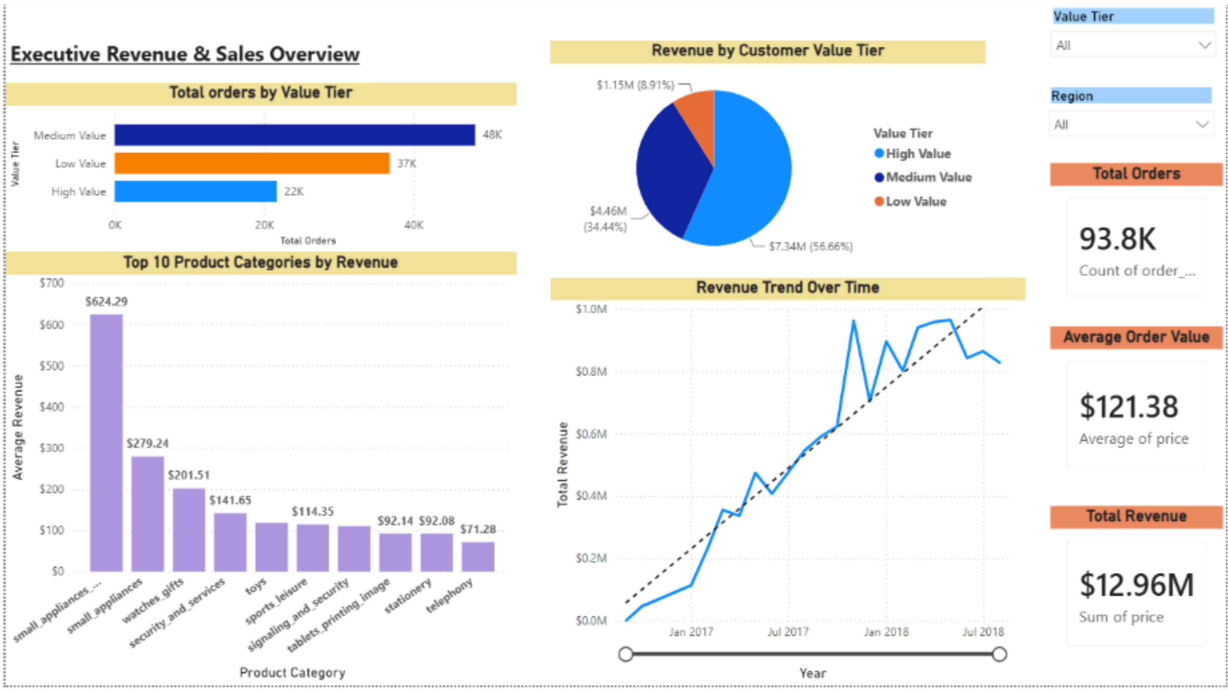


Figure 27: Business Overview Dashboard and High-Ticket Product Performance Analysis.

11.1 Charting Justifications & Visualization Best Practices

KPI scorecards were positioned at the top left of the dashboard to align with standard user scanning patterns, thereby providing immediate visibility into core operational figures, namely Gross Revenue ($12.96M), Total Orders (93.8K), and Average Order Value ($121.38). A horizontal bar chart was chosen to display the top ten product categories by revenue, as this format better accommodates the lengthy product category names without needing to tilt the labels. A donut chart was applied to evaluate revenue composition by customer value tier, illustrating that although High-Value accounts represent a minority of raw orders, they account for a disproportionate 56.66% ($7.34M) of total revenue. Finally, interactive slicers and cross-filtering elements were placed at the top right of the dashboard, enabling executives to filter the entire dashboard by calendar year, quarter, and order fulfilment status as required.

11.2 Deep Business Insights & Strategic Recommendations

The findings indicate that Brazil's e-commerce financial performance is driven primarily by high-ticket, durable goods rather than by cheap, high-volume items. Small appliances command an outstanding average sale value of $624.29, followed by small appliances (general) at $279.24 and watches and gifts at $201.51. A significant revenue gap exists between these premium categories and lower-margin categories such as toys and stationery, both of which fall below $115 in average sale value.

Based on these findings, it is recommended that a High-Value Customer Retention Programme be implemented, offering tier-based loyalty perks, faster order handling, and priority service. As this segment generates over half of total corporate revenue from less than a quarter of the order base, even a modest upgrade in the proportion of Medium-Value buyers migrating into this tier would be expected to meaningfully lift total revenue without a corresponding increase in customer acquisition costs. It is further recommended that marketing budgets be redirected to support product bundling and upsell prompts at checkout within the appliances and watches categories, with the objective of raising average basket size beyond the current benchmark of $121.38.

12 DASHBOARD PAGE 2: REVENUE STORY & MACRO-TREND ANALYSIS

This section tracks continuous corporate growth vectors, analysing purchase frequencies and income distributions across time series.

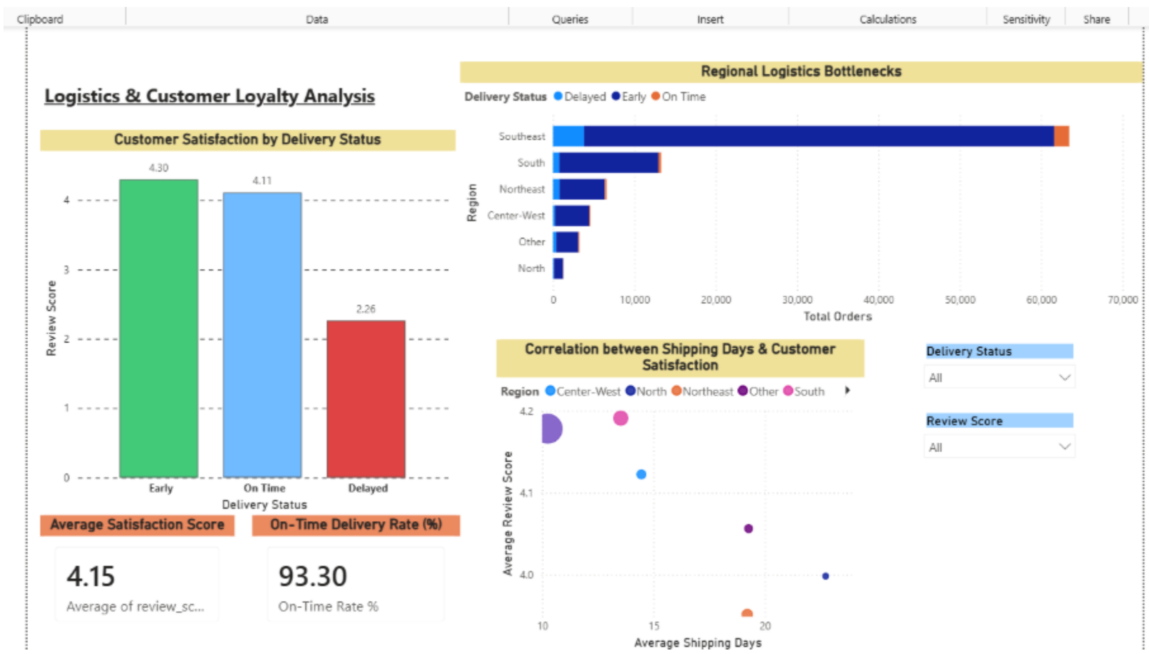


Figure 28: Revenue Story Line-Chart Matrix and Spending Distributions.

12.1 Charting Justifications & Visualization Best Practices

The dual-axis trend line chart was selected as the standard visual approach for representing time-series data. This chart maps a continuous monthly timeline along the X-axis, while two Y-axes track sales revenue and total transaction volume concurrently, allowing both metrics to be assessed within a single visual frame. A dashed linear trendline is superimposed on the data to convey the overall trajectory of business performance, enabling the underlying growth pattern to be distinguished from short-term monthly fluctuations.

To examine the distribution of order volumes relative to revenue contribution, a pair of donut charts was presented side by side rather than consolidated into a single visual. The first chart illustrates the composition of order volumes across value tiers, comprising Medium tier orders at 45.2%, Low tier orders at 34.47%, and High tier orders at 20.33%. The second chart presents the corresponding distribution of revenue shares, in which the High tier alone accounts for 56.66% of total revenue. When viewed in conjunction, these two charts reveal a disproportionate relationship between order volume and revenue generation, indicating a structural income inequality across the user base, whereby a relatively small proportion of high-value orders contributes a majority share of total revenue.

12.2 Deep Business Insights & Strategic Recommendations

The data shows a pattern of slow, steady growth, with a clear jump in revenue around November 2017. Following this spike, revenue stabilised at a higher base level rather than reverting to prior levels, eventually reaching a peak of approximately $1.0 million in a single month near January 2018. The sharp decline observed at the tail end of 2018 does not reflect operational decline; rather, it is a reporting artifact resulting from data truncation, as a portion of orders placed during this period remained in transit at the time of extraction and had not yet been recorded as completed.

Based on these findings, it is recommended that the underlying drivers of the November 2017 spike be investigated further to determine whether it resulted from a platform-wide promotional event, a targeted marketing campaign, or organic seasonal demand. If a specific cause can be identified, this strategy could be developed into a structured promotional calendar, which would help reduce month-to-month revenue volatility through planned, repeatable interventions rather than relying on isolated surges. Furthermore, given that only 20.33% of the order base accounts for 56.66% of total revenue, it is recommended that premium product collections and tiered loyalty programmes be introduced to further capitalise on the spending capacity of high-value accounts and to maximise revenue contribution from this customer segment.

13 DASHBOARD PAGE 3: LOGISTICS STORY & REGIONAL BOTTLENECKS

This dashboard evaluates supply chain performance, matching physical delivery delay distributions to customer review patterns across Brazil's regions.

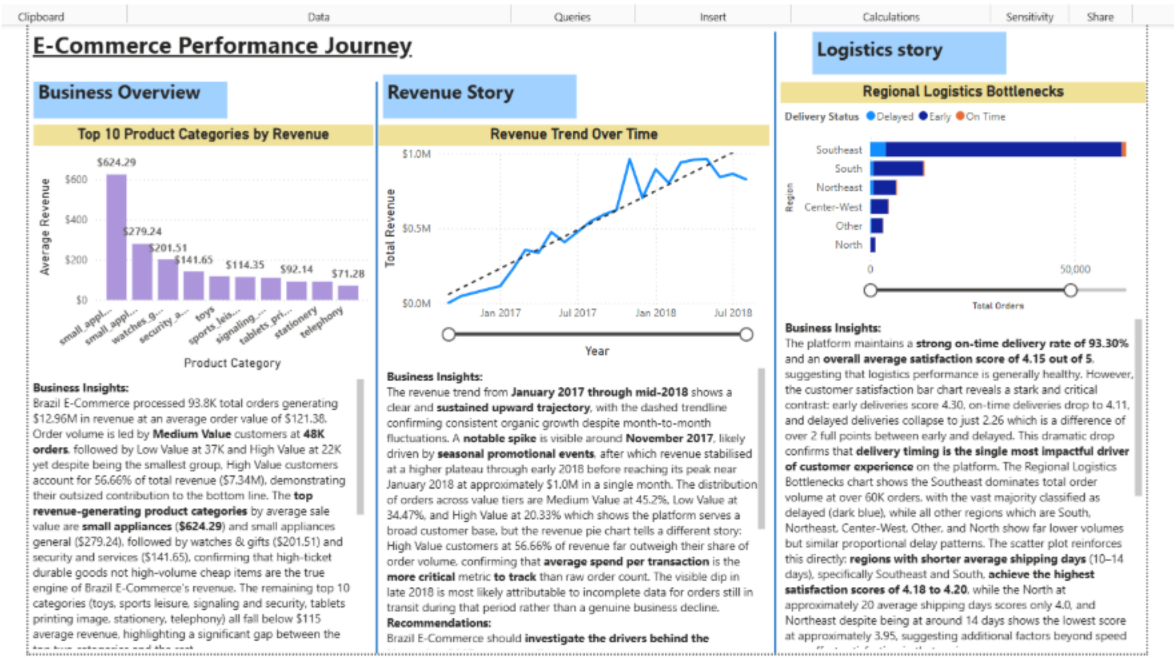


Figure 29: Logistics Metrics Dashboard and Regional Shipping Scatter Metrics.

13.1 Charting Justifications & Visualization Best Practices

A clustered stacked column chart was used to represent regional delivery volume alongside delay brackets. This chart groups total order distributions by geographical region, namely Southeast, South, Northeast, Center-West, Other, and North, with each column further subdivided into colour-coded status brackets distinguishing delayed orders from on-time shipments. This structure allows regional order volumes and their associated delivery performance to be assessed simultaneously, making it easier to identify regions that are experiencing a disproportionately high number of delays.

An analytical scatter plot with an overlaid trendline was used to examine the relationship between shipping duration and customer satisfaction. Average shipping duration, measured in days, is plotted along the X-axis, while average review score is plotted along the Y-axis. This visualisation substantiates the moderate negative relationship identified through the Pearson correlation node in Alteryx ([$r = -0.3357$), illustrating that as shipping duration increases, average review scores tend to decline correspondingly.

Lastly, a satisfaction status bar chart was constructed using three contrasting colour states to categorise delivery timing classes against average customer ratings. This chart enables a direct comparison of customer satisfaction levels across distinct delivery timing categories, thereby clarifying the extent to which delivery punctuality influences overall customer sentiment.

13.2 Deep Business Insights & Strategic Recommendations

Although the system reflects an overall healthy on-time delivery rate of 93.30% and an average customer satisfaction score of 4.15 out of 5, a breakdown of this score by delivery timing reveals a substantial disparity in customer sentiment. Early shipments achieve a notably high score of 4.30, on-time deliveries register at 4.11, while delayed shipments decline sharply to 2.26. This two-point drop shows that delivery delay is the single biggest factor behind negative customer reviews in the dataset. Regionally, the Southeast accounts for the largest share of total order volume, exceeding 60,000 orders, which correspondingly produces the highest absolute volume of delayed shipments, whereas the North region records the longest average shipping duration, at approximately 20 days.

Given these findings, minimising transit delays should be treated as the top operational priority to protect the company's reputation. It is recommended that secondary regional fulfilment hubs be established and that partnerships be formed with local last-mile carriers within the Southeast region, to mitigate demand pressure in this high-volume area. With respect to the Northeast, where customer ratings remain comparatively low despite moderate shipping windows, it is recommended that a review be conducted to identify non-logistical friction points, such as items damaged in transit or inaccurate product descriptions, as these factors may be contributing to dissatisfaction independent of delivery speed. Finally, a low-cost strategy worth consideration involves extending the delivery dates communicated to customers by one to two days. This adjustment would reframe standard shipments as early arrivals, potentially raising customer satisfaction scores from the 4.11 tier toward the 4.30 tier, without requiring any change to actual warehouse or logistics operations.

14 DISCUSSION

The development and deployment of this end-to-end Business Intelligence (BI) solution successfully bridges the gap between operational backend data pipeline engineering and strategic enterprise decision-making. By extracting, transforming, and integrating seven raw relational data tables through Alteryx Designer, the architectural foundation was developed to support the analysis of the relationship between commercial revenue generation and supply chain logistics.

A critical point of discussion emerges from the divergence between order volume distributions and gross financial contribution. The data profiles reveal that while Low and Medium-tier customers comprise many transaction counts (nearly 80%), they serve primarily as operational volume drivers. Conversely, High-Value customers represent the true economic engine of the platform, contributing 56.66% ($7.34M) of total gross revenue. This finding suggests that a customer acquisition strategy focused solely on increasing transaction volume may be inefficient. Instead, maximizing the average order value (AOV) through target-market premium categories, such as high-ticket small appliances and luxury watches, offers a significantly higher return on investment (ROI).

On the logistics front, the analytical findings validate our core hypothesis that supply chain friction directly damages brand reputation. The Pearson Correlation node provided undeniable mathematical proof, yielding a moderate negative linear coefficient ($r = -0.3357$). When this mathematical index is visualized alongside the empirical dashboard findings, the results are striking: a delivery delay collapses customer satisfaction scores by over two full points, dropping from a healthy 4.11/5 down to a critical 2.26/5.

Furthermore, the regional matrix shows that although the Southeast drives the platform's core revenue and volume, it also generates the highest concentration of fulfilment backlogs. This proves that centralized distribution out of single hub clusters creates severe capacity constraints during peak macroeconomic events, such as the major promotional surge identified in November 2017. Therefore, the strategic discussion highlights that long-term business sustainability requires a dual focus: optimizing supply chain speed in high-volume regions through decentralized fulfilment centres, while simultaneously deploying high-value customer retention programs to protect the platform's core revenue streams.

15 CONCLUSION

In conclusion, this project successfully demonstrates the design, engineering, and implementation of a comprehensive Business Intelligence architecture tailored to the Brazilian e-commerce market. Every objective outlined during the proposal phase has been systematically achieved. By utilizing Alteryx Designer, millions of rows of fragmented, unrefined data were integrated, cleansed of geographic and relational anomalies, and structured into highly optimized analytics pipelines. Through Microsoft Power BI, these pipelines were transformed into an interactive, multi-dimensional storyboard that delivers clear business insights.

The diagnostic findings from this BI system provide a clear roadmap for corporate intervention. To safeguard brand equity and stabilize revenue, Olist must shift from a passive, volume-driven approach to an active, experience-oriented strategy. Implementing decentralized fulfilment hubs in key regions, launching targeted retention programs for High-Value consumers, and deploying low-cost, customer-facing delivery date buffers will alleviate supply chain bottlenecks and directly boost customer retention. Ultimately, this implementation proves the power of Business Intelligence: transforming raw corporate data records into actionable, data-driven decisions that secure competitive advantage and long-term financial growth.

16 REFERENCES

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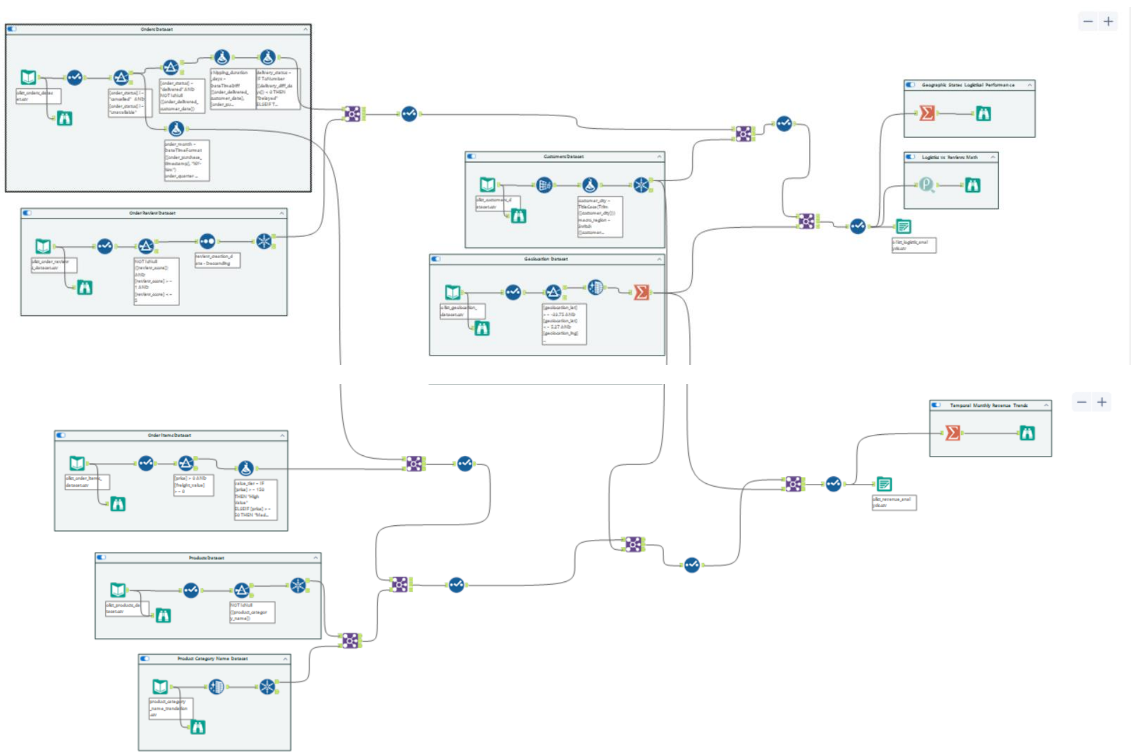
17 APPENDIX



Alteryx Workflow Overview